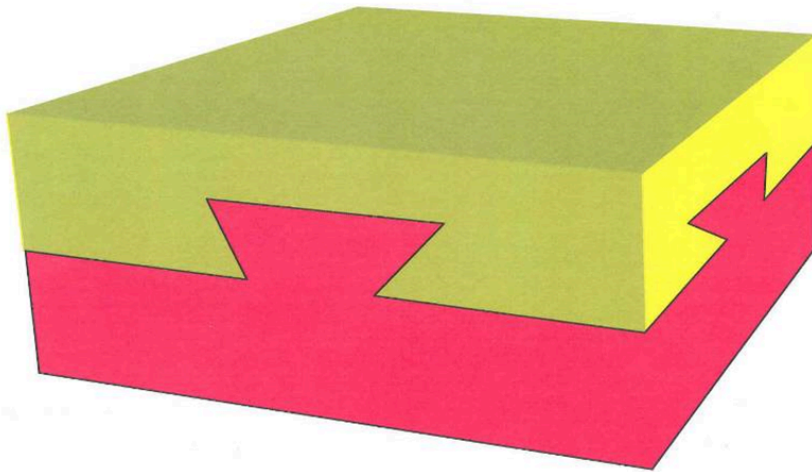


The Unnamed 3d Thingy

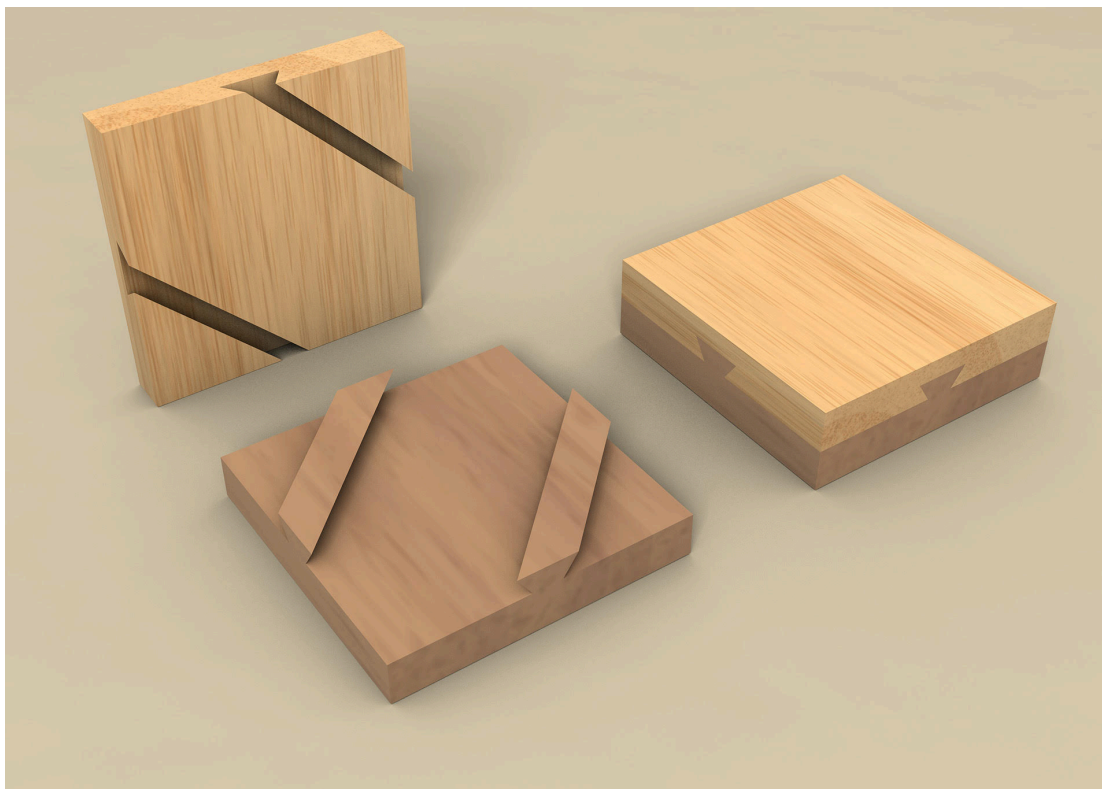
Years ago, my boss shared with me a book he wrote, in digital format; its main topic is critical thinking training. There is a section about a mental exercise, that I am summarizing as follows:

There is a block composed of two different types of wood, each about an inch in height, that were mated in a manner that ensured no gaps of air between them. From every side, the assembly looks the same:



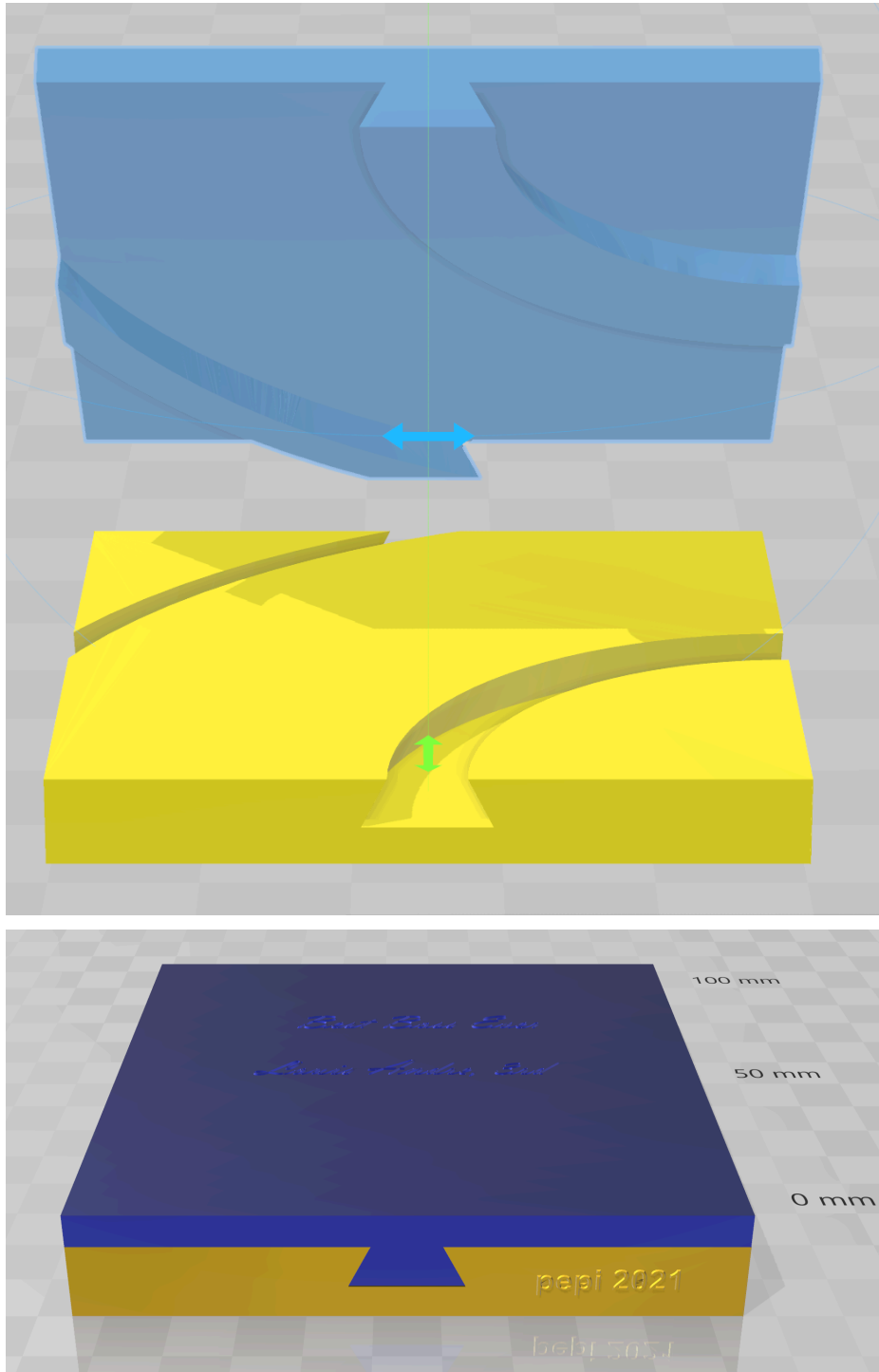
The two layers of wood can be separated from each other by a single simple movement, there is no glue or any other adhesive or mechanical inhibitor involved. They could be reunited with the same type of motion.

In the book, the solution to this puzzle is exactly the one beautifully designed by Roger:



(I really love it!)

My observation back then, which I rummaged a bit more over time, is that this is not *the solution*, because in fact the problem admits infinite solutions. To prove it, I created a different one using 3D Builder on Windows, I had it 3D printed and gave it to my boss around Christmas as a “semi-parting” gift, as he is semi-retiring:



As it can be easily verified, Roger's solution and mine belong to a family of solutions which consist of two grooves each connecting one of the side keys to another, where the center of the curve that they describe can be arbitrarily set from somewhere between the middle of the square and one of the corners, then to a corner (as in my solution), then the beyond the corner all the way to infinity (Roger's solution).

There are of course infinite solutions in this family, but they are still very much limited in their shape; in particular, there is no additional information that can be encoded in the shape itself.

After giving it a bit of thought, I found out that we can do much better than that.

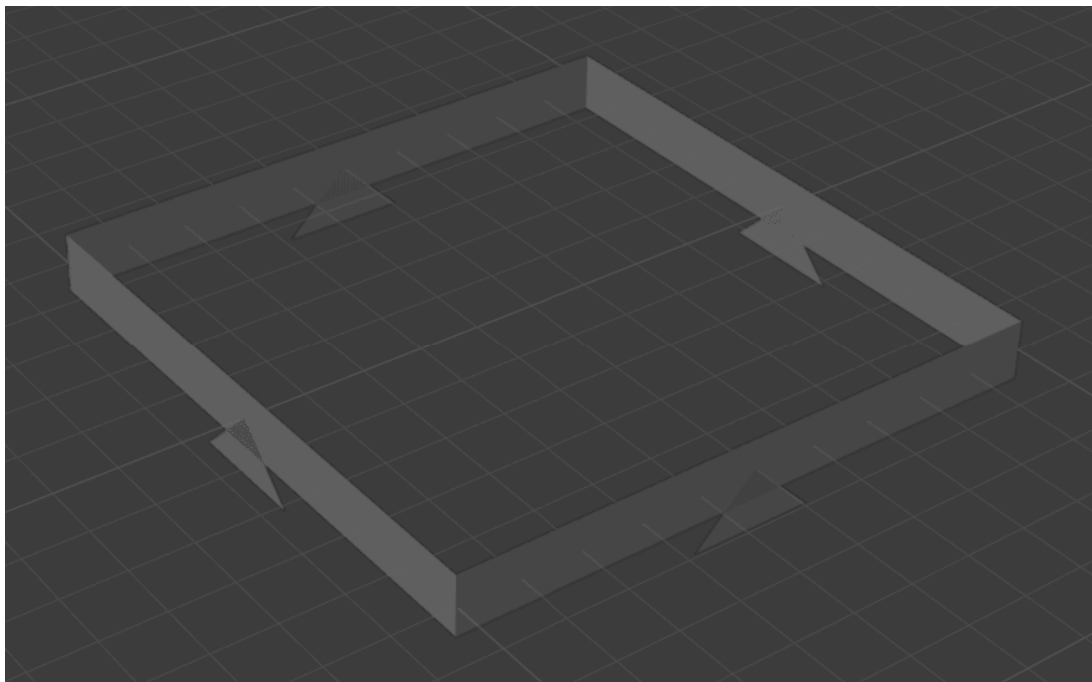
In fact, any amount of information, such as, for instance, QR codes or Roger's portrait, can be encoded *inside* the apparatus without violating the requirements. Furthermore, the method can be extended from four to any number of sides ≥ 3 .

In a nutshell, the process is as follow:

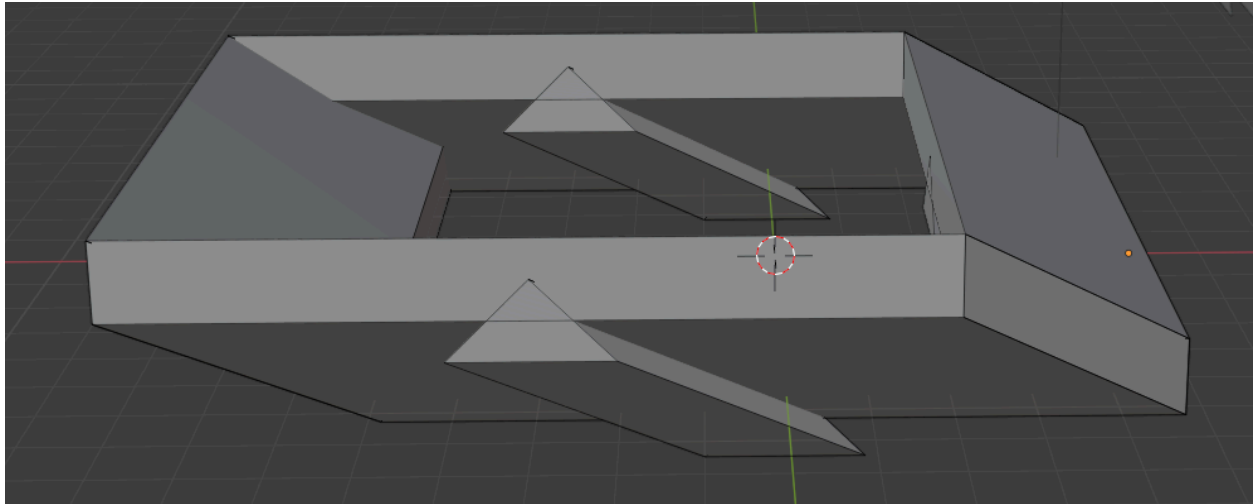
- define a base shape for the half side



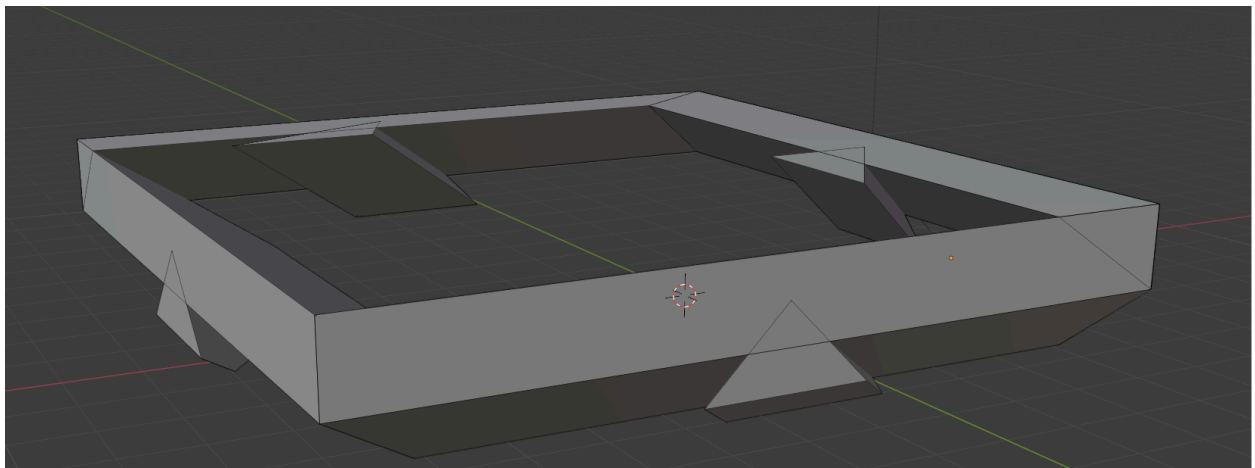
- mirror it to generate a full side
- compose the sides to build the "polygon box" of choice, a square in this example but of course it can be anything ≥ 3



- extrude the box along a line with constant curvature, which can be arbitrary with some constraints, e.g. it cannot lie on a plane that is parallel to any of the sides. In this example, the extrusion is carried out along a straight line that is not parallel to any of the axes of the square prism, but it could be any other from an infinity of possible curves



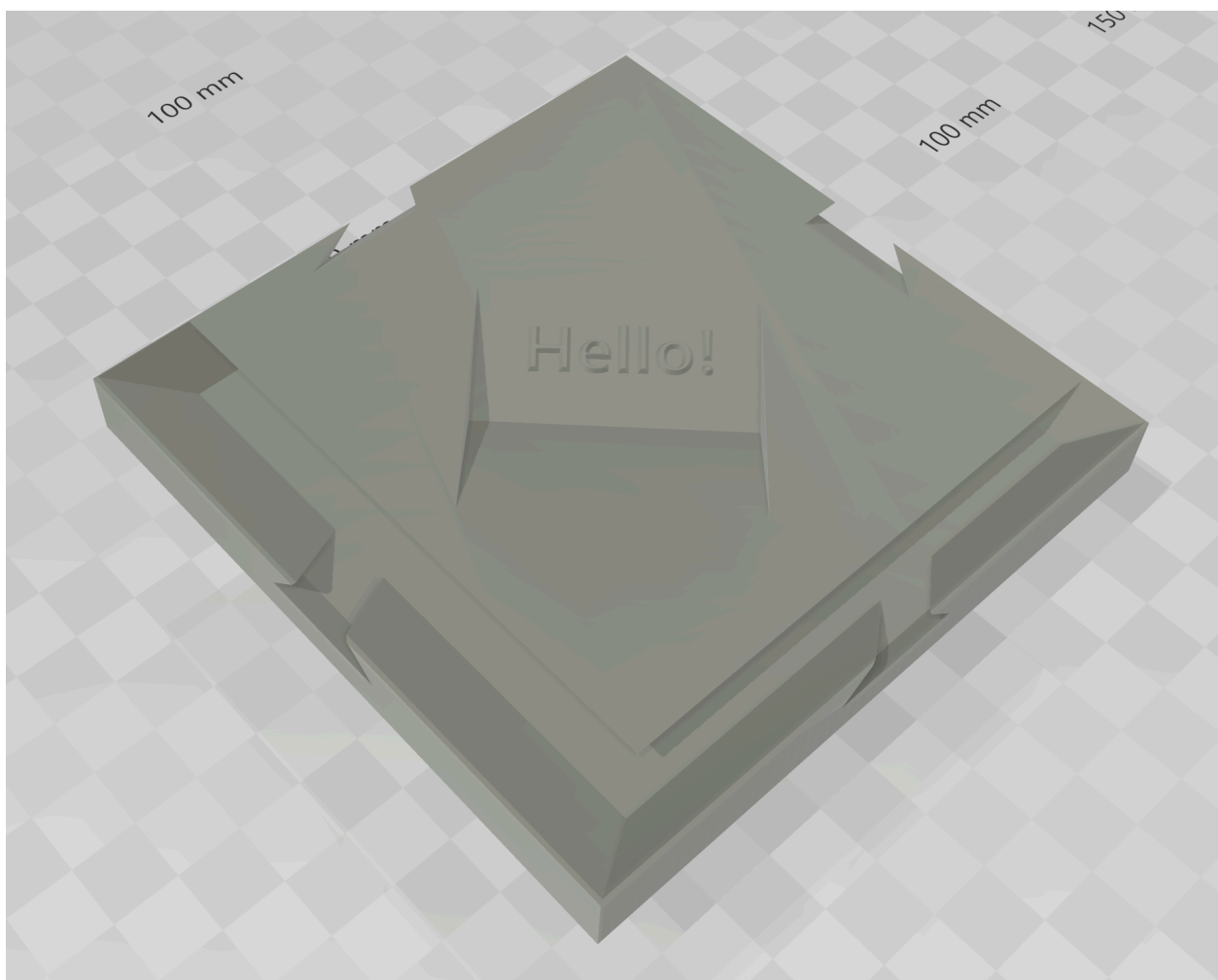
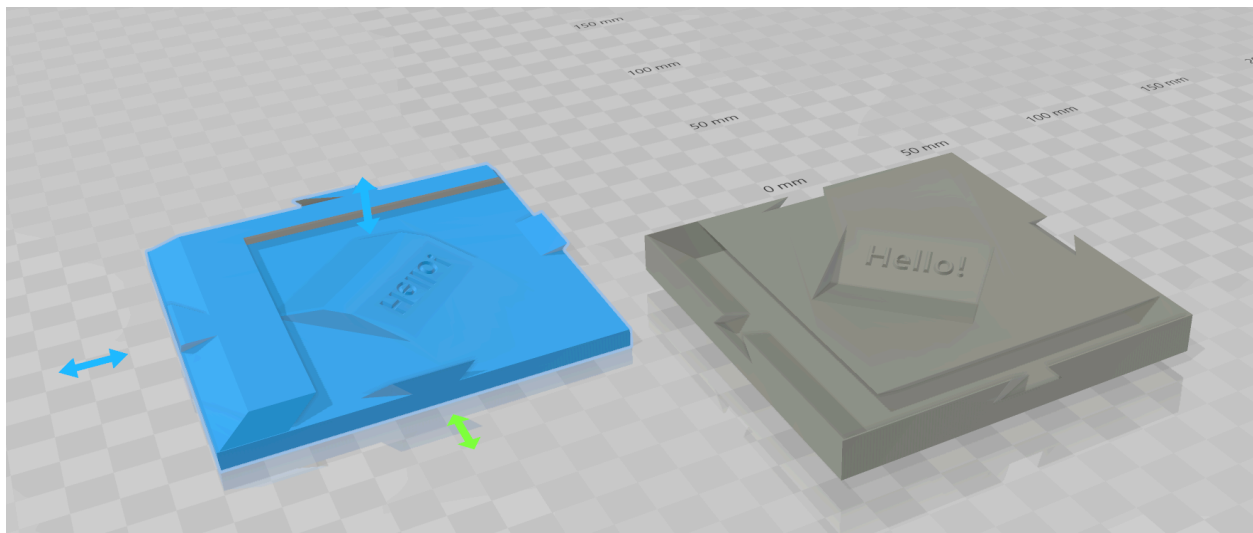
- compose the 3d boundary box for the complete assembly, that is, a prism that matches the contour of the "polygon box" and extends to surround the volume of the missing element
- crop the extrusion using the boundary box



Now, all the mandatory features are implemented; all we are left to do is to place a “lid” on top of the box, and then create the complementing part by filling the space below. That means that we still have a lot of degrees of freedom in designing the rest of the boundary between the two parts, and thus we could encode an arbitrary amount of information there.

The key difference between this method and the solutions mentioned at the beginning of the document is that, by moving along three axes instead of two, we were able to give each side its own independent groove, which in turn makes it possible to change arbitrarily the number of sides.

Putting it all together, I built a quick complete example using Blender up to the step above, and then assembling it with 3D Builder, using .stl for the exchange:



A graduation U3T ("Unnamed 3D Thingy") for Francesco

Based on the above, we have a little project for the upcoming graduation in the family - I know, it sounds impossible! I wonder, could it be reused elsewhere? ;)

The basic idea: starting from the "Hello" design in the previous page, adding two enhancements:

- Carrying out the extrusion of the "polygon box" along a nice, creative curve instead of a predictable and boring straight line
- Maximize use of the available space in the box, including the following elements:
- A nice title "Francesco Strafforello"
- Subtitle "University of Toronto '22"
- QR code (will point to a family album online)
- (only if feasible) other cool geometric elements TBD
- To account for engineering tolerance, we create two slightly different matrices, inner and outer (I'd rather stay out of gender references)

1 - matrix

The dimensions of the final product will be mm 100x100x15

Construct using 3D Builder

Tolerance guidance (from Xometry handbook):

GENERAL MACHINING TOLERANCES

For features of size (L, W, H, $\varnothing$) and location (position, concentricity, symmetry):
 $\pm 0.005"$

Sharp edges will be broken and deburred by default. Critical edges that must be left sharp should be noted or specified on a print.

For features of orientation (parallelism $\parallel$ and perpendicularity $\perp$) and form (cylindrical, flatness ∇ , circularity $\bigcirc$, and straightness —) apply tolerances as follows:

Distance Length (inches)	Bilateral Tolerances (+ inches)	Angularity degrees
0-12"	$\pm 0.005"$	$\pm \frac{1}{2}^\circ$
12-34"	$\pm 0.010"$	$\pm \frac{1}{2}^\circ$
24-36"	$\pm 0.016"$ (1/64")	$\pm 1^\circ$
36-60"	$\pm 0.031"$ (1/32")	$\pm 1^\circ$
Over 60"	$\pm 0.063"$ (1/16")	$\pm 1^\circ$

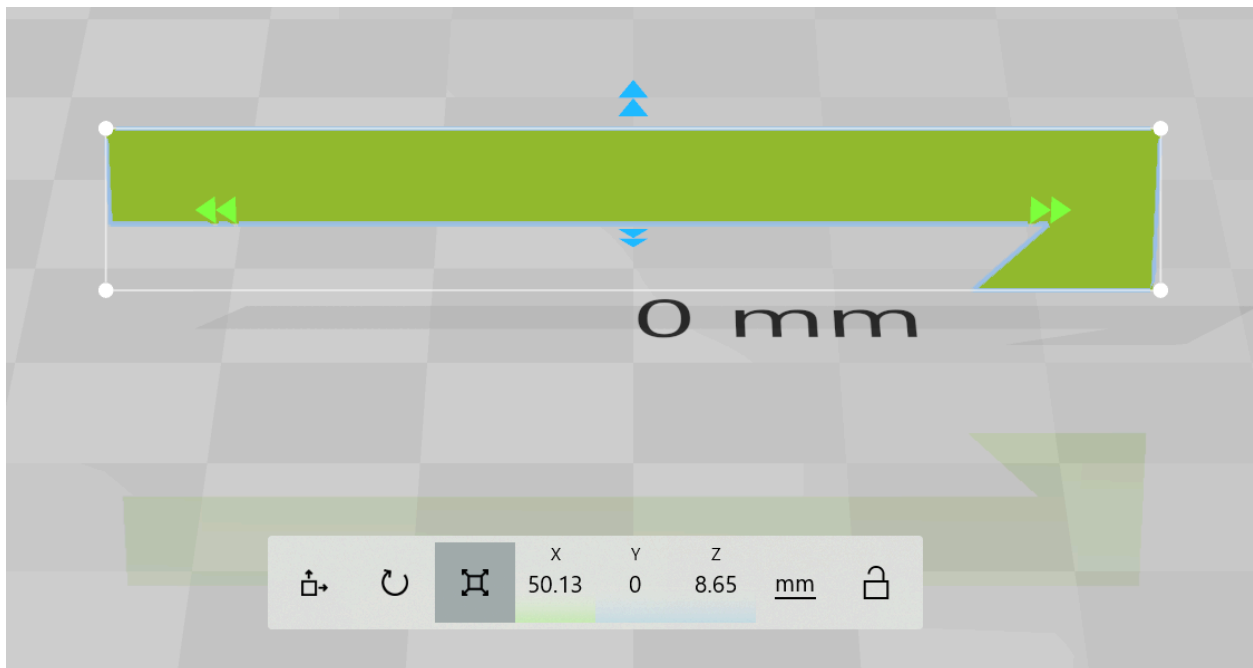
Given the dimensions of our project, it seems that applying a tolerance of $\pm 0.005" = 0.127 \text{ mm}$ everywhere might work.

To introduce tolerance, we modify the construction process described above. While we were going through the extrusion process for one of the mating parts, and filling the remaining space to obtain the other one, now we are going to extrude the tolerance space, and then using the free space on either side to obtain the actual parts. With that in mind, we start by creating the matrix as the difference between the inner and the outer one.

Inner matrix:

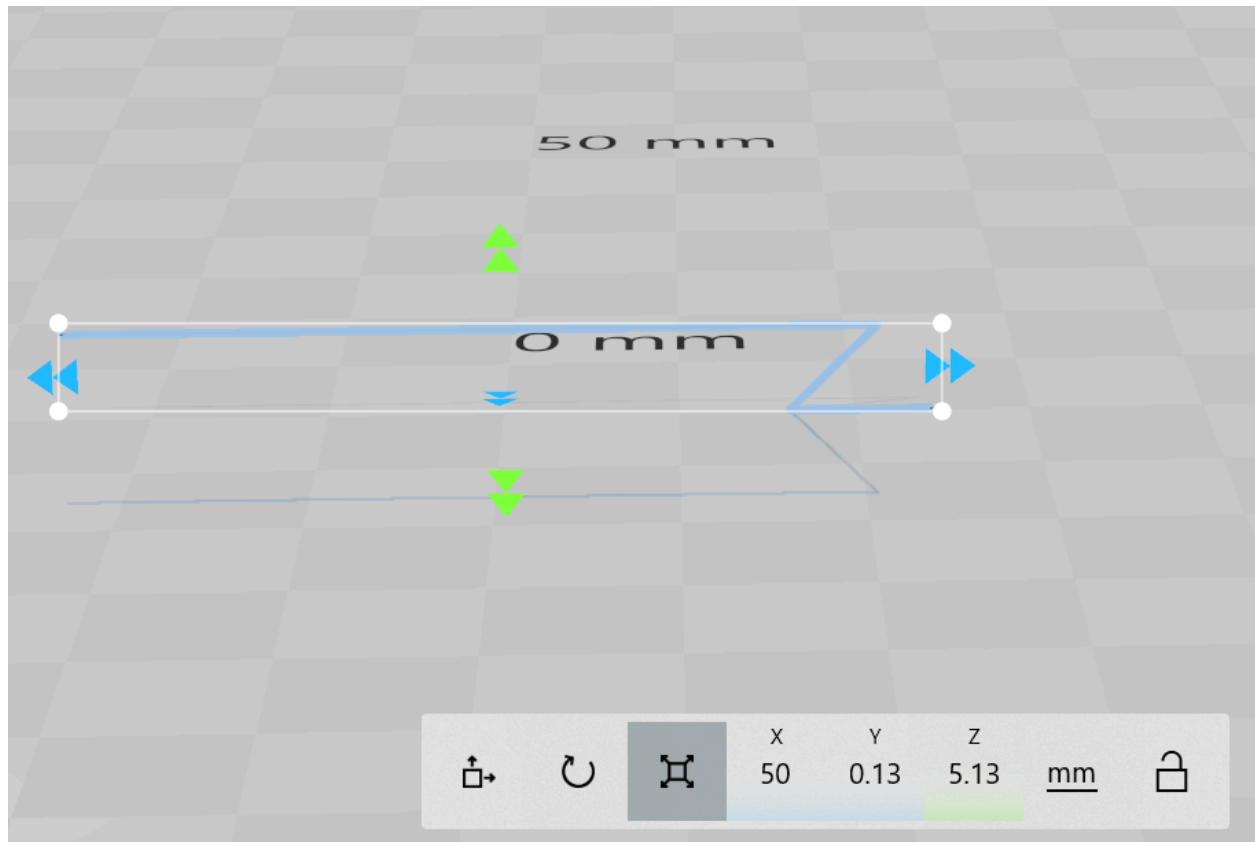


Outer matrix:

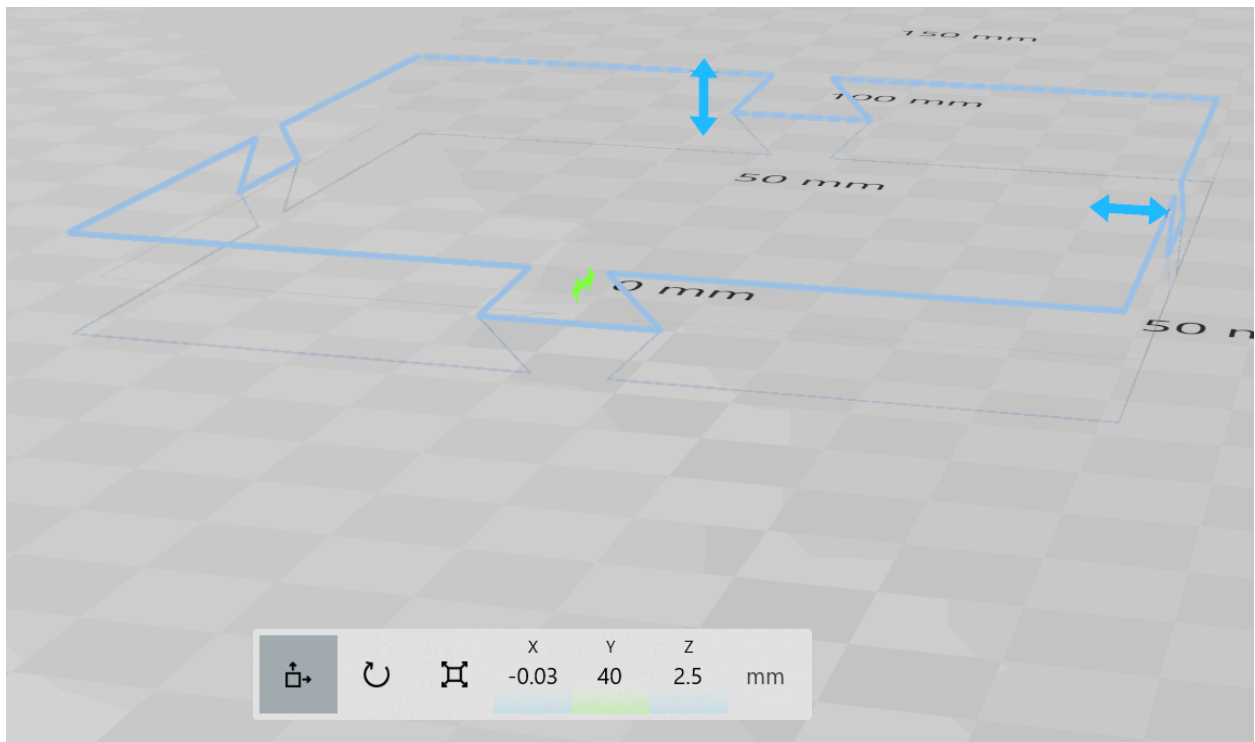


The difference in the longer side (0.15 instead of 0.127) is adopted to make extra room for diagonals and various curves.

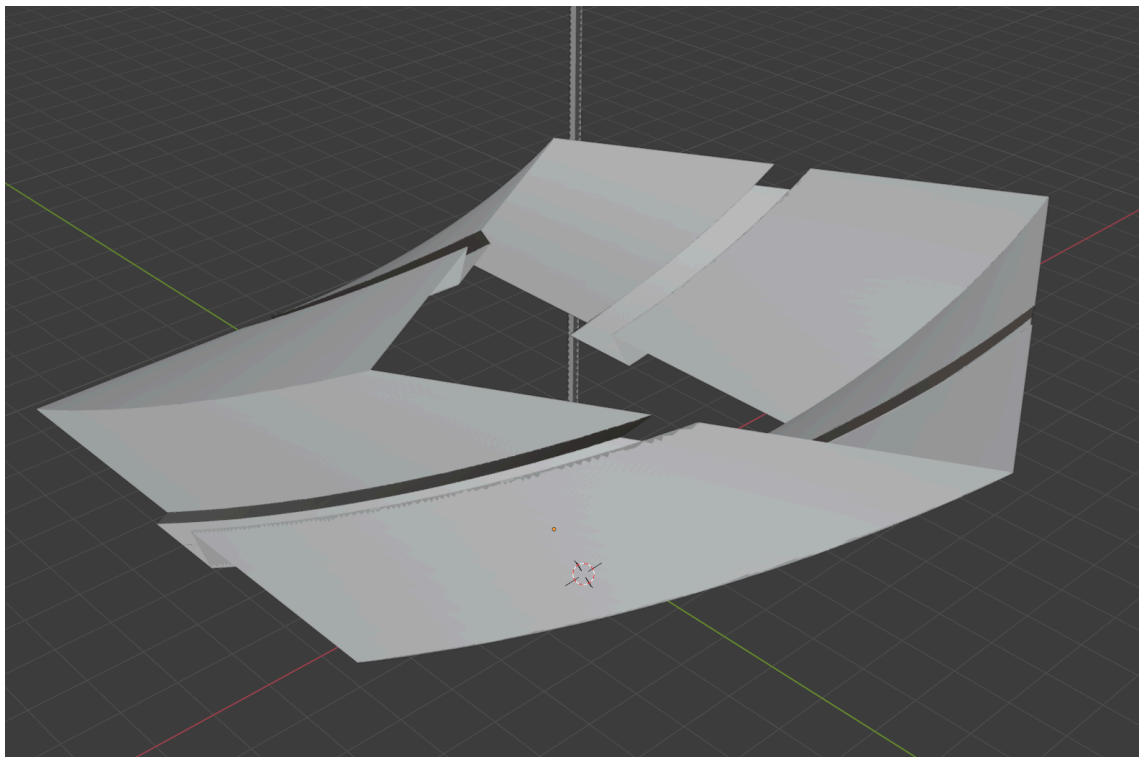
In the end, the new matrix is simply composed of three thin rods joined together:



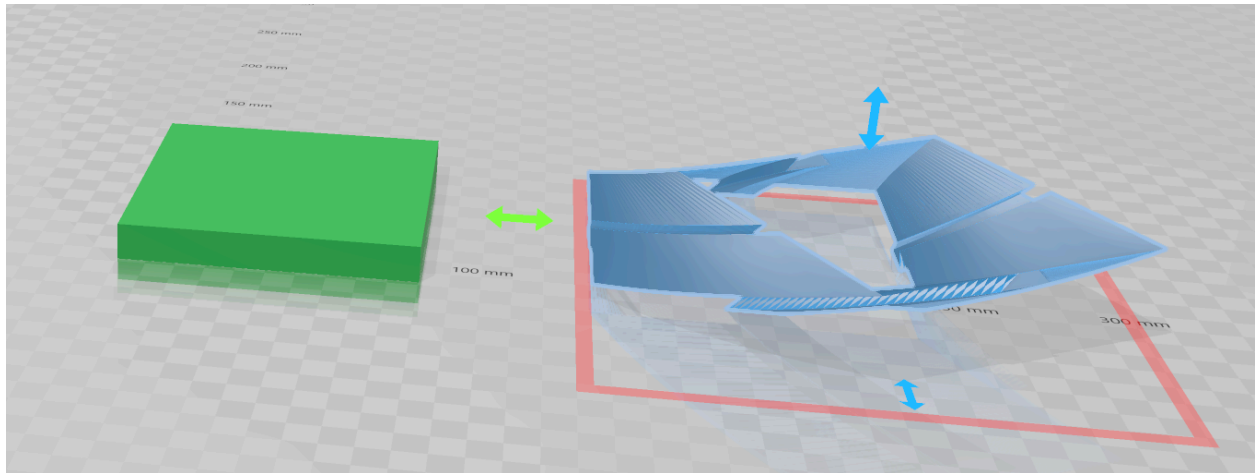
And this is our “polygon box” - that is, the apparatus before the extrusion:



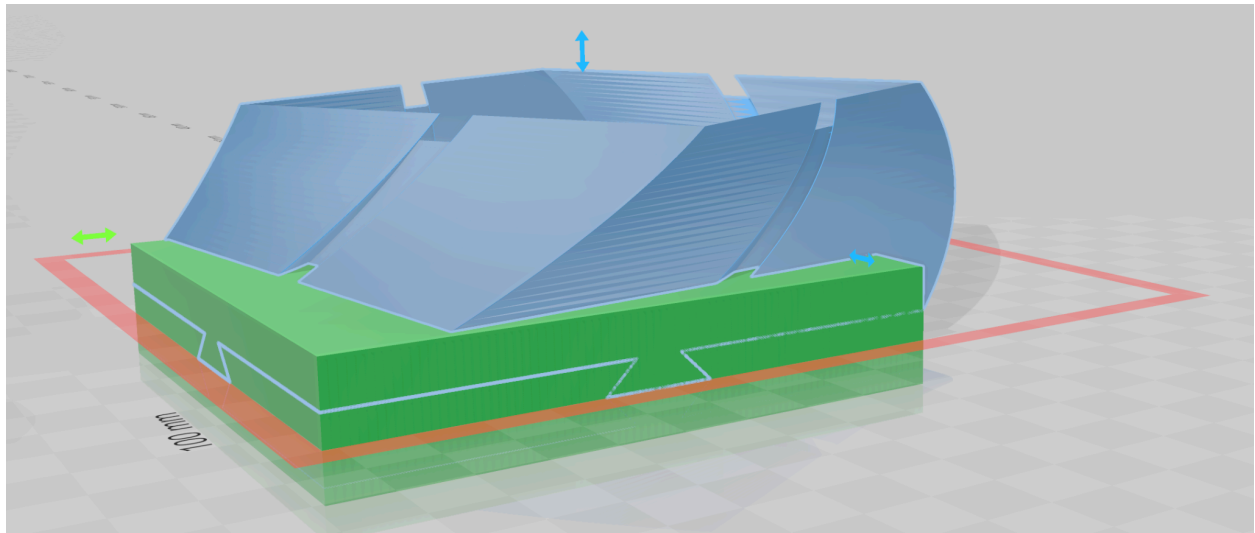
Thus far, I have used 3D builder. Now I export the model in .STL format, to perform the extrusion with Blender. After a few attempts, I found that rotating the polygon box and applying the screw modifier along an external object I could get a nice, curved surface (external object is the elongated cuboid in the background):



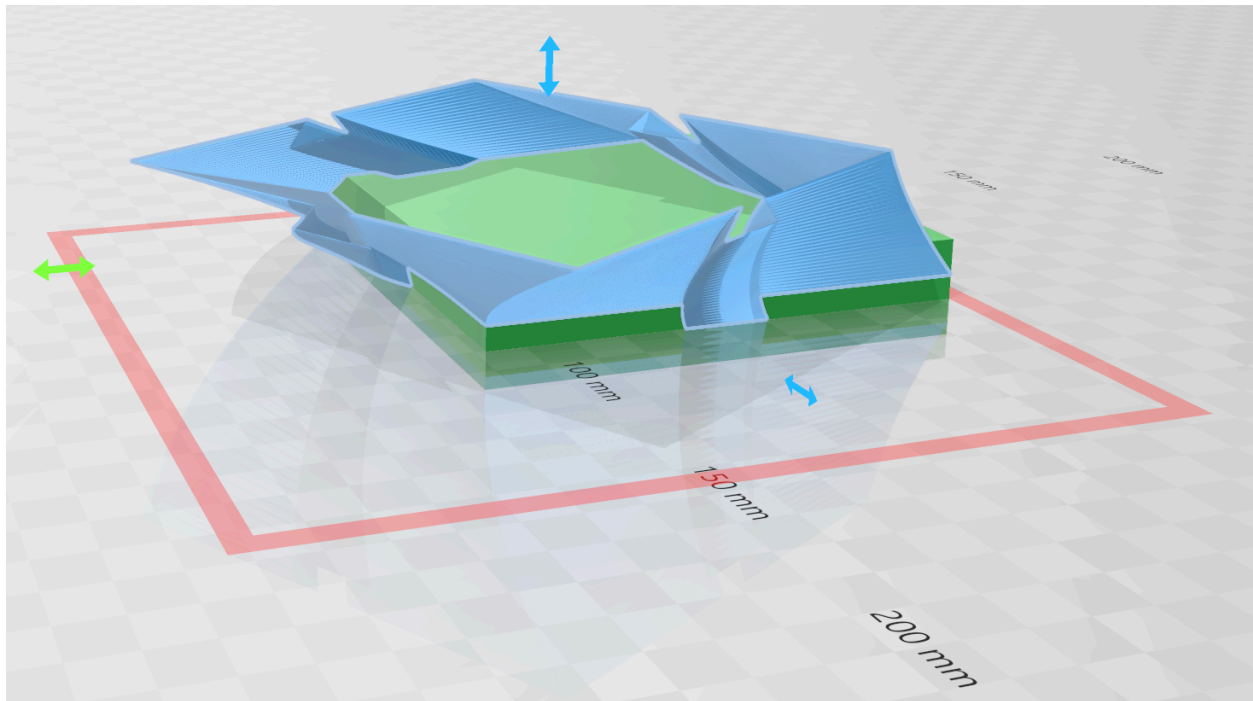
After exporting the component to .stl, and importing it back to Builder 3D we can obtain this scene:



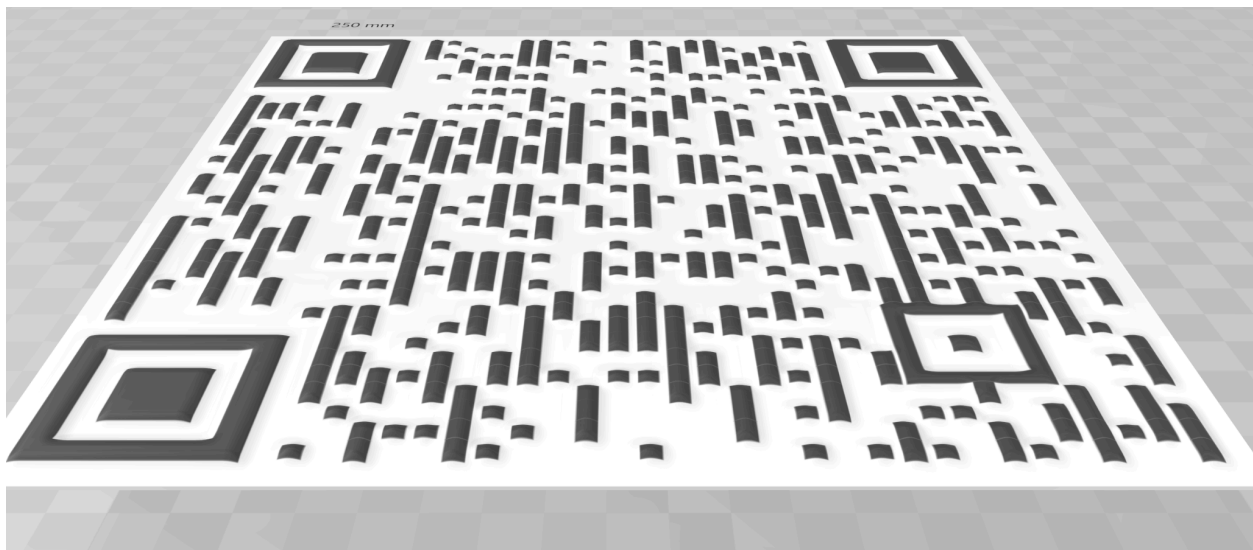
The green cuboid is the final volume for the apparatus, the blue one is the imported artifact. Note the red square around it, something in its geometry seems to be problematic for 3D builder; but, activating the “Repair” function actually alters its geometry. I’m trying to go ahead as it is, everything seems to be working so far. Checking out the intersection of the two artifacts:



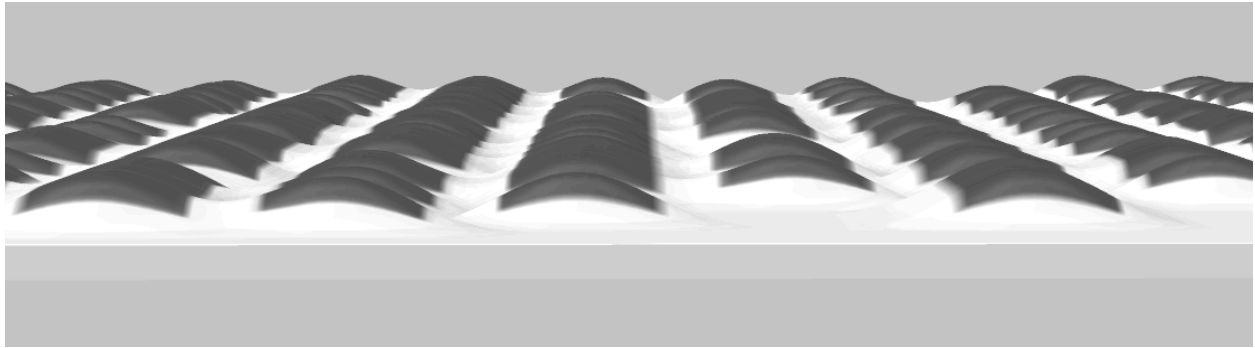
Everything seem to be correct from the opposite side as well:



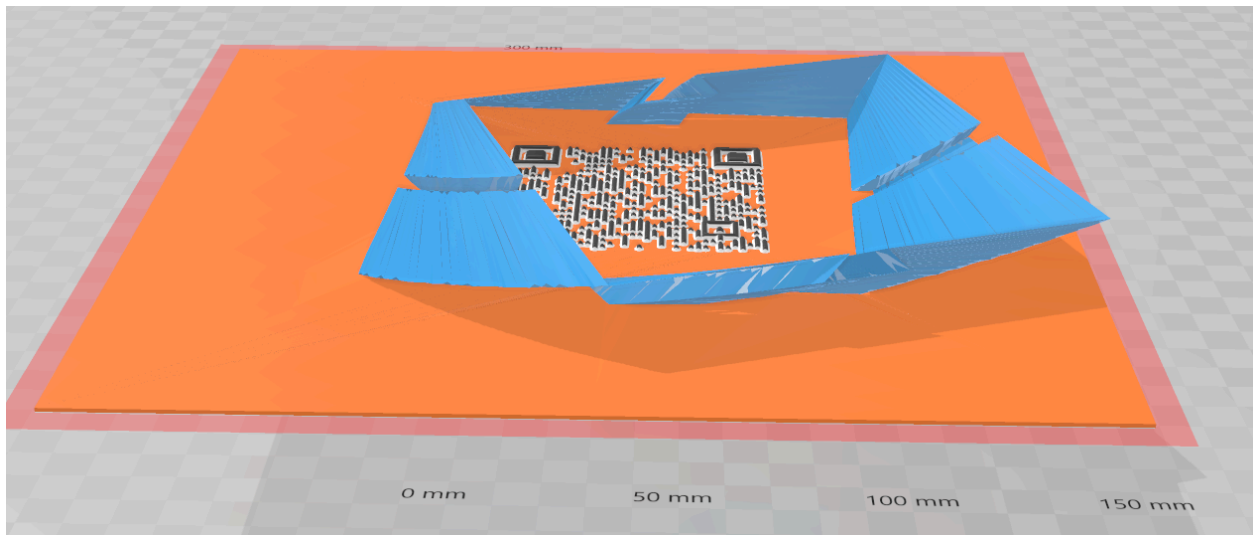
Next step: I need to add a bottom to my “cutting box” - the blue shape, that is. The bottom is where I want to add some hidden information - in this case, a QR code pointing to Francesco’s LinkedIn profile. I generated the QR code using QRmonkey.org, but I could have used the sharing feature in Chrome, etc., then I loaded in 3D Builder and gave it a bit of 3D relevance (see <https://www.youtube.com/watch?v=plvNbWHi84Y>)



I am making sure that the angles of the 3D features in the bottom QR-code image need to be flatter than whatever angles will come from the “cutting box”:

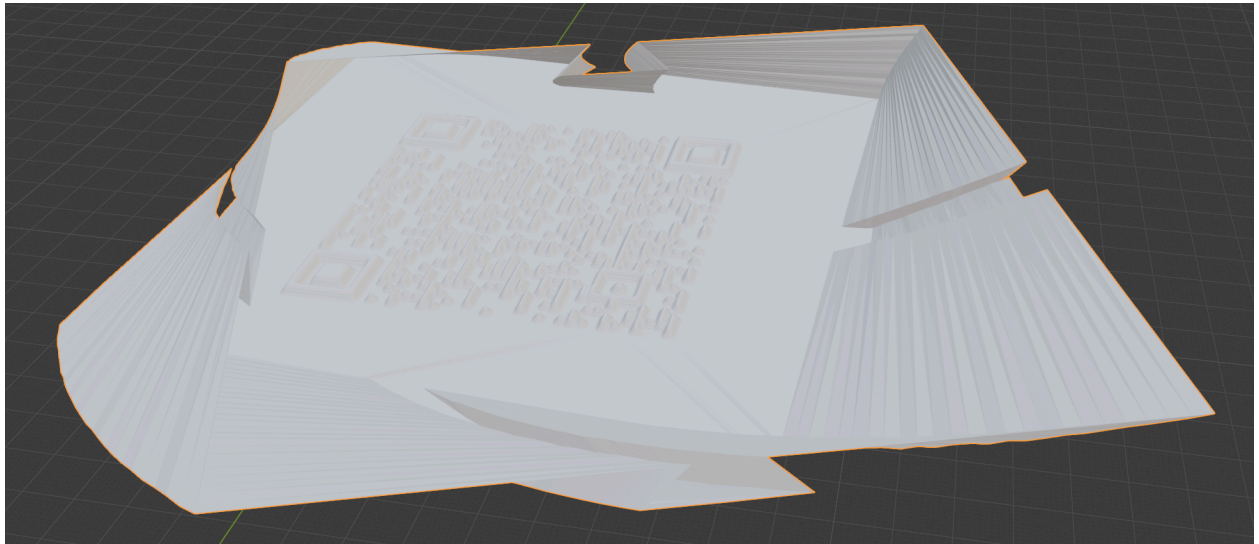


Working inside 3D Builder I was able to join the bottom with the QR code and the cutting artifact, but I could not manage to cleanup the material in excess from the bottom:



So I exported the shape to STL and loaded it in Blender.

Then, with some hours of mesh work in Blender, I completed the cutting box which now includes the bottom with QR code:



Since this shape has been entirely generated using a base shape that has a slightly larger width than the known material tolerance, I can hope that removing it from the base box I will get two mating parts; of course, the mesh is always an approximation but I can hope that in the worst case I'll have to sand them a little.

OK, I'm saving this one just because I think it is cool:

